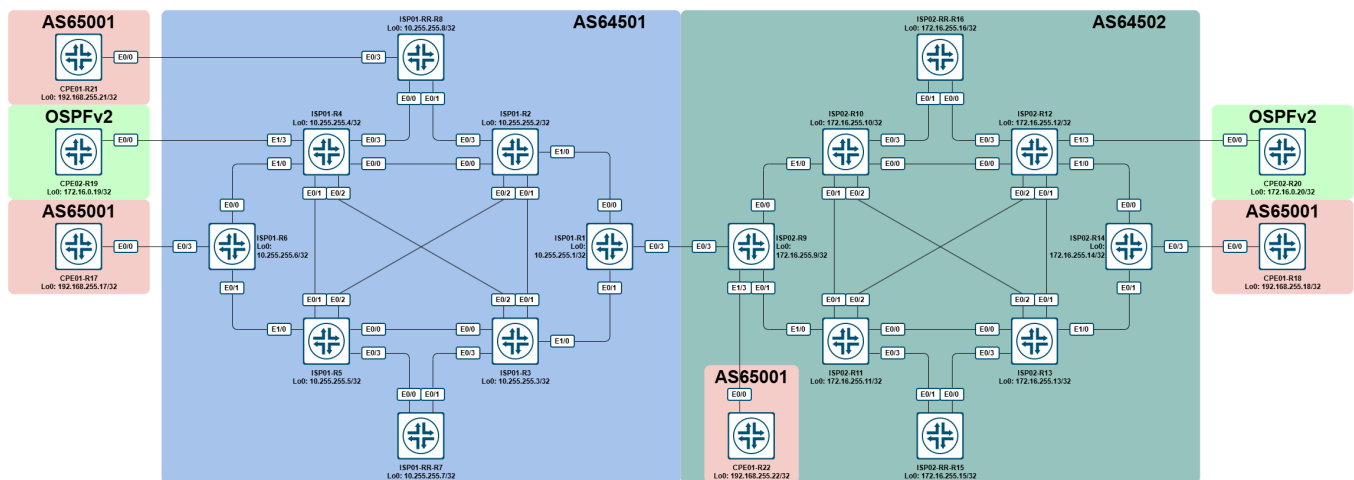


LAB - Cisco - BGP Option C MPLS L3VPN

- LAB - Cisco - BGP Option C MPLS L3VPN
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Топология



Описание

В данной лабораторной работе необходимо обеспечить связность между сайтами клиентов, подключенных к разным автономным системам. Для этого необходимо организовать interAS-стык по схеме «option C».

Данную задачу можно разбить на две последовательные подзадачи:

1. Поднятие interAS-стыка в AFI/SAFI IPv4 Labeled-unicast между ASBR'ами.
2. Поднятие interAS-стыков в AFI/SAFI VPNv4 Unicast между рефлекторами, расположенных в разных AS.

Настройка

Настройка - IPv4 Labeled Unicast

AS64501:

R1:

```
router bgp 64501
 address-family ipv4
  network 10.255.255.1 mask 255.255.255.255
```

```
neighbor 10.255.255.7 activate
neighbor 10.255.255.8 activate
neighbor RR send-community
neighbor RR send-label explicit-null
exit-address-family
!
!
```

R2:

```
router bgp 64501
address-family ipv4
network 10.255.255.2 mask 255.255.255.255
neighbor 10.255.255.7 activate
neighbor 10.255.255.8 activate
neighbor RR send-community
neighbor RR send-label explicit-null
exit-address-family
!
!
```

R3:

```
router bgp 64501
address-family ipv4
network 10.255.255.3 mask 255.255.255.255
neighbor 10.255.255.7 activate
neighbor 10.255.255.8 activate
neighbor RR send-community
neighbor RR send-label explicit-null
exit-address-family
!
!
```

R4:

```
router bgp 64501
address-family ipv4
network 10.255.255.4 mask 255.255.255.255
neighbor 10.255.255.7 activate
neighbor 10.255.255.8 activate
neighbor RR send-community
neighbor RR send-label explicit-null
exit-address-family
!
!
```

R5:

```
router bgp 64501
 address-family ipv4
  network 10.255.255.5 mask 255.255.255.255
  neighbor 10.255.255.7 activate
  neighbor 10.255.255.8 activate
  neighbor RR send-community
  neighbor RR send-label explicit-null
 exit-address-family
!
```

R6:

```
router bgp 64501
 address-family ipv4
  network 10.255.255.6 mask 255.255.255.255
  neighbor 10.255.255.7 activate
  neighbor 10.255.255.8 activate
  neighbor RR send-community
  neighbor RR send-label explicit-null
 exit-address-family
!
```

R7:

```
router bgp 64501
!
 address-family ipv4
  network 10.255.255.7 mask 255.255.255.255
  neighbor 10.255.255.1 activate
  neighbor 10.255.255.2 activate
  neighbor 10.255.255.3 activate
  neighbor 10.255.255.4 activate
  neighbor 10.255.255.5 activate
  neighbor 10.255.255.6 activate
  neighbor 10.255.255.8 activate
  neighbor OWN_RRC send-community
  neighbor OWN_RRC send-label explicit-null
  neighbor OWN_RRC route-reflector-client
  neighbor RRN send-community
  neighbor RRN send-label explicit-null
 exit-address-family
!
```

R8:

```

router bgp 64501
!
address-family ipv4
 network 10.255.255.8 mask 255.255.255.255
 neighbor 10.255.255.1 activate
 neighbor 10.255.255.2 activate
 neighbor 10.255.255.3 activate
 neighbor 10.255.255.4 activate
 neighbor 10.255.255.5 activate
 neighbor 10.255.255.6 activate
 neighbor 10.255.255.7 activate
 neighbor OWN_RRC send-community
 neighbor OWN_RRC send-label explicit-null
 neighbor OWN_RRC route-reflector-client
 neighbor RRN send-community
 neighbor RRN send-label explicit-null
exit-address-family
!
!

```

Проверяем в AS64501:

R8:

```

ISP01-RR-R8#show bgp ipv4 unicast | b Network

```

	Network	Next Hop	Metric	LocPrf	Weight	Path
r i	10.255.255.1/32	10.255.255.1	0	100	0	i
r>i		10.255.255.1	0	100	0	i
r i	10.255.255.2/32	10.255.255.2	0	100	0	i
r>i		10.255.255.2	0	100	0	i
r i	10.255.255.3/32	10.255.255.3	0	100	0	i
r>i		10.255.255.3	0	100	0	i
r i	10.255.255.4/32	10.255.255.4	0	100	0	i
r>i		10.255.255.4	0	100	0	i
r i	10.255.255.5/32	10.255.255.5	0	100	0	i
r>i		10.255.255.5	0	100	0	i
r i	10.255.255.6/32	10.255.255.6	0	100	0	i
r>i		10.255.255.6	0	100	0	i
r>i	10.255.255.7/32	10.255.255.7	0	100	0	i
*>	10.255.255.8/32	0.0.0.0	0		32768	i

```

ISP01-RR-R8#show bgp ipv4 unicast labels

```

Network	Next Hop	In label/Out label
10.255.255.1/32	10.255.255.1	nolabel/exp-null
	10.255.255.1	nolabel/exp-null
10.255.255.2/32	10.255.255.2	nolabel/exp-null
	10.255.255.2	nolabel/exp-null
10.255.255.3/32	10.255.255.3	nolabel/exp-null
	10.255.255.3	nolabel/exp-null
10.255.255.4/32	10.255.255.4	nolabel/exp-null

	10.255.255.4	nolabel/exp-null
10.255.255.5/32	10.255.255.5	nolabel/exp-null
	10.255.255.5	nolabel/exp-null
10.255.255.6/32	10.255.255.6	nolabel/exp-null
	10.255.255.6	nolabel/exp-null
10.255.255.7/32	10.255.255.7	nolabel/exp-null
10.255.255.8/32	0.0.0.0	imp-null/nolabel

AS64502:**R09:**

```
router bgp 64502
 address-family ipv4
  network 172.16.255.9 mask 255.255.255.255
  neighbor 172.16.255.15 activate
  neighbor 172.16.255.16 activate
  neighbor RR send-community
  neighbor RR send-label explicit-null
 exit-address-family
!
```

R10:

```
router bgp 64502
 address-family ipv4
  network 172.16.255.10 mask 255.255.255.255
  neighbor 172.16.255.15 activate
  neighbor 172.16.255.16 activate
  neighbor RR send-community
  neighbor RR send-label explicit-null
 exit-address-family
!
```

R11:

```
router bgp 64502
 address-family ipv4
  network 172.16.255.11 mask 255.255.255.255
  neighbor 172.16.255.15 activate
  neighbor 172.16.255.16 activate
  neighbor RR send-community
  neighbor RR send-label explicit-null
 exit-address-family
```

```
!  
!
```

R12:

```
router bgp 64502  
  address-family ipv4  
    network 172.16.255.12 mask 255.255.255.255  
    neighbor 172.16.255.15 activate  
    neighbor 172.16.255.16 activate  
    neighbor RR send-community  
    neighbor RR send-label explicit-null  
  exit-address-family  
!  
!
```

R13:

```
router bgp 64502  
  address-family ipv4  
    network 172.16.255.13 mask 255.255.255.255  
    neighbor 172.16.255.15 activate  
    neighbor 172.16.255.16 activate  
    neighbor RR send-community  
    neighbor RR send-label explicit-null  
  exit-address-family  
!  
!
```

R14:

```
router bgp 64502  
  address-family ipv4  
    network 172.16.255.14 mask 255.255.255.255  
    neighbor 172.16.255.15 activate  
    neighbor 172.16.255.16 activate  
    neighbor RR send-community  
    neighbor RR send-label explicit-null  
  exit-address-family  
!  
!
```

R15:

```

router bgp 64502
!
address-family ipv4
network 172.16.255.15 mask 255.255.255.255
neighbor 172.16.255.9 activate
neighbor 172.16.255.10 activate
neighbor 172.16.255.11 activate
neighbor 172.16.255.12 activate
neighbor 172.16.255.13 activate
neighbor 172.16.255.14 activate
neighbor OWN_RRC send-community
neighbor OWN_RRC send-label explicit-null
neighbor OWN_RRC route-reflector-client
exit-address-family
!
!

```

R16:

```

router bgp 64502
!
address-family ipv4
network 172.16.255.16 mask 255.255.255.255
neighbor 172.16.255.9 activate
neighbor 172.16.255.10 activate
neighbor 172.16.255.11 activate
neighbor 172.16.255.12 activate
neighbor 172.16.255.13 activate
neighbor 172.16.255.14 activate
neighbor OWN_RRC send-community
neighbor OWN_RRC send-label explicit-null
neighbor OWN_RRC route-reflector-client
exit-address-family
!
!

```

Проверяем в AS64502:

R15:

```

ISP02-RR-R15#show bgp ipv4 unicast | b Network

```

Network	Next Hop	Metric	LocPrf	Weight	Path
r>i 172.16.255.9/32	172.16.255.9	0	100	0	i
r>i 172.16.255.10/32	172.16.255.10	0	100	0	i
r>i 172.16.255.11/32	172.16.255.11	0	100	0	i
r>i 172.16.255.12/32	172.16.255.12	0	100	0	i
r>i 172.16.255.13/32	172.16.255.13	0	100	0	i
r>i 172.16.255.14/32	172.16.255.14	0	100	0	i

```
*> 172.16.255.15/32 0.0.0.0 0 32768 i
ISP02-RR-R15#show bgp ipv4 unicast labels
  Network          Next Hop      In label/Out label
172.16.255.9/32    172.16.255.9    nolabel/exp-null
172.16.255.10/32   172.16.255.10   nolabel/exp-null
172.16.255.11/32   172.16.255.11   nolabel/exp-null
172.16.255.12/32   172.16.255.12   nolabel/exp-null
172.16.255.13/32   172.16.255.13   nolabel/exp-null
172.16.255.14/32   172.16.255.14   nolabel/exp-null
172.16.255.15/32   0.0.0.0         imp-null/nolabel
```

Сейчас поднимем BGP-стык между ASBR'ми:

R1:

```
router bgp 64501
!
neighbor 192.0.2.1 remote-as 64502
neighbor 192.0.2.1 update-source Ethernet0/3
!
address-family ipv4
neighbor 192.0.2.1 activate
neighbor 192.0.2.1 send-community
neighbor 192.0.2.1 send-label explicit-null
neighbor RR next-hop-self
!
!
```

R9:

```
router bgp 64502
!
neighbor 192.0.2.0 remote-as 64501
neighbor 192.0.2.0 update-source Ethernet0/3
!
address-family ipv4
neighbor 192.0.2.0 activate
neighbor 192.0.2.0 send-community
neighbor 192.0.2.0 send-label explicit-null
neighbor RR next-hop-self
!
!
```

Как и в случае с optB, при поднятии BGP-сессии, подразумевающей обмен метками и прохождение MPLS-пакетов, включается MPLS на интерфейсе:

R1:

```

ISP01-ASBR-R1#
*Mar 19 13:18:45.522: %BGP-5-ADJCHANGE: neighbor 192.0.2.1 Up
ISP01-ASBR-R1#
*Mar 19 13:18:45.525: %BGP_LMM-6-AUTOGEN1: The mpls bgp forwarding command has
been configured on interface: Ethernet0/3
ISP01-ASBR-R1#show mpls interface
Interface          IP          Tunnel    BGP Static Operational
Ethernet0/0        Yes (ldp)   No        No  No    Yes
Ethernet0/1        Yes (ldp)   No        No  No    Yes
Ethernet0/3        No          No        Yes No    Yes
ISP01-ASBR-R1#show ip route 192.0.2.0
Routing entry for 192.0.2.0/24, 3 known subnets
  Attached (3 connections)
    Variably subnetted with 2 masks
C       192.0.2.0/31 is directly connected, Ethernet0/3
L       192.0.2.0/32 is directly connected, Ethernet0/3
C       192.0.2.1/32 is directly connected, Ethernet0/3
ISP01-ASBR-R1#

```

Проверяем RIB и LFIB на ASBR:

R1:

```

ISP01-ASBR-R1#show ip route bgp | b Gateway
Gateway of last resort is not set

    172.16.0.0/32 is subnetted, 8 subnets
B       172.16.255.9 [20/0] via 192.0.2.1, 00:06:26
B       172.16.255.10 [20/0] via 192.0.2.1, 00:06:26
B       172.16.255.11 [20/0] via 192.0.2.1, 00:06:26
B       172.16.255.12 [20/0] via 192.0.2.1, 00:06:26
B       172.16.255.13 [20/0] via 192.0.2.1, 00:06:26
B       172.16.255.14 [20/0] via 192.0.2.1, 00:06:26
B       172.16.255.15 [20/0] via 192.0.2.1, 00:06:26
B       172.16.255.16 [20/0] via 192.0.2.1, 00:06:26
ISP01-ASBR-R1#show mpls forwarding-table next-hop 192.0.2.1
Local      Outgoing  Prefix          Bytes Label  Outgoing  Next Hop
Label      Label     or Tunnel Id   Switched     interface
10000      Pop Label  192.0.2.1/32   0            Et0/3      192.0.2.1
10001      explicit-n 172.16.255.9/32 0            Et0/3      192.0.2.1
10002      90009     172.16.255.10/32 0            Et0/3      192.0.2.1
10011      90008     172.16.255.11/32 0            Et0/3      192.0.2.1
10012      90007     172.16.255.12/32 0            Et0/3      192.0.2.1
10013      90006     172.16.255.13/32 0            Et0/3      192.0.2.1
10014      90005     172.16.255.14/32 0            Et0/3      192.0.2.1
10015      90004     172.16.255.15/32 0            Et0/3      192.0.2.1
10016      90003     172.16.255.16/32 0            Et0/3      192.0.2.1
ISP01-ASBR-R1#

```

Проверяем связность:

R1:

```
ISP01-ASBR-R1#ping 172.16.255.9 source 10.255.255.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.255.9, timeout is 2 seconds:
Packet sent with a source address of 10.255.255.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
ISP01-ASBR-R1#ping 172.16.255.15 source 10.255.255.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.255.15, timeout is 2 seconds:
Packet sent with a source address of 10.255.255.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

Теперь посмотрим тоже самое на RR:

R8:

```
ISP01-RR-R8#show ip route bgp | b Gateway
Gateway of last resort is not set

172.16.0.0/32 is subnetted, 8 subnets
B       172.16.255.9 [200/0] via 10.255.255.1, 00:16:01
B       172.16.255.10 [200/0] via 10.255.255.1, 00:16:01
B       172.16.255.11 [200/0] via 10.255.255.1, 00:16:01
B       172.16.255.12 [200/0] via 10.255.255.1, 00:16:01
B       172.16.255.13 [200/0] via 10.255.255.1, 00:16:01
B       172.16.255.14 [200/0] via 10.255.255.1, 00:16:01
B       172.16.255.15 [200/0] via 10.255.255.1, 00:16:01
B       172.16.255.16 [200/0] via 10.255.255.1, 00:16:01
ISP01-RR-R8#show mpls forwarding-table next-hop 10.255.255.1
Local      Outgoing  Prefix          Bytes Label  Outgoing  Next Hop
Label      Label      or Tunnel Id    Switched     interface
ISP01-RR-R8#show mpls forwarding-table
Local      Outgoing  Prefix          Bytes Label  Outgoing  Next Hop
Label      Label      or Tunnel Id    Switched     interface
80002      explicit-n 10.255.255.4/32  0            Et0/0      10.4.8.4
80003      40008      10.255.255.6/32  0            Et0/0      10.4.8.4
80004      40006      10.255.255.5/32  0            Et0/0      10.4.8.4
80005      20009      10.255.255.7/32  0            Et0/1      10.2.8.2
            40011      10.255.255.7/32  0            Et0/0      10.4.8.4
80006      explicit-n 10.255.255.2/32  0            Et0/1      10.2.8.2
80007      20005      10.255.255.1/32  0            Et0/1      10.2.8.2
80008      20006      10.255.255.3/32  0            Et0/1      10.2.8.2
80009      Pop Label  10.0.0.8/32[V]   0            aggregate/MGMT
80010      No Label   10.8.21.0/24[V]  0            aggregate/CUST1
80011      No Label   192.168.255.21/32[V]  \
```

ISP01-RR-R8#	0	Et0/3	10.8.21.21
--------------	---	-------	------------

В RIB маршруты есть, а вот в LFIB, как ни странно, соответствия меток и префиксов нужных нет. Посмотрим, что у нас видно в BGP:

R8:

```
ISP01-RR-R8#show bgp ipv4 unicast labels
```

Network	Next Hop	In label/Out label
10.255.255.1/32	10.255.255.1	nolabel/exp-null
	10.255.255.1	nolabel/exp-null
10.255.255.2/32	10.255.255.2	nolabel/exp-null
	10.255.255.2	nolabel/exp-null
10.255.255.3/32	10.255.255.3	nolabel/exp-null
	10.255.255.3	nolabel/exp-null
10.255.255.4/32	10.255.255.4	nolabel/exp-null
	10.255.255.4	nolabel/exp-null
10.255.255.5/32	10.255.255.5	nolabel/exp-null
	10.255.255.5	nolabel/exp-null
10.255.255.6/32	10.255.255.6	nolabel/exp-null
	10.255.255.6	nolabel/exp-null
10.255.255.7/32	10.255.255.7	nolabel/exp-null
10.255.255.8/32	0.0.0.0	imp-null/nolabel
172.16.255.9/32	10.255.255.1	nolabel/10001
	10.255.255.1	nolabel/10001
172.16.255.10/32	10.255.255.1	nolabel/10002
	10.255.255.1	nolabel/10002
172.16.255.11/32	10.255.255.1	nolabel/10011
	10.255.255.1	nolabel/10011
172.16.255.12/32	10.255.255.1	nolabel/10012
	10.255.255.1	nolabel/10012
172.16.255.13/32	10.255.255.1	nolabel/10013
	10.255.255.1	nolabel/10013
172.16.255.14/32	10.255.255.1	nolabel/10014
	10.255.255.1	nolabel/10014
172.16.255.15/32	10.255.255.1	nolabel/10015
	10.255.255.1	nolabel/10015
172.16.255.16/32	10.255.255.1	nolabel/10016
	10.255.255.1	nolabel/10016

Здесь мы видим, что метки для лупбэков из AS64502 получены. Посмотрим более детально на какой-нибудь из префиксов:

R8:

```
ISP01-RR-R8#show bgp ipv4 unicast 172.16.255.9/32
BGP routing table entry for 172.16.255.9/32, version 10
Paths: (2 available, best #2, table default)
```

```
Advertised to update-groups:
  1          2
Refresh Epoch 1
64502
  10.255.255.1 (metric 200) from 10.255.255.7 (10.255.255.7)
    Origin IGP, metric 0, localpref 100, valid, internal
    Originator: 10.255.255.1, Cluster list: 10.255.255.7
    mpls labels in/out nlabel/10001
    rx pathid: 0, tx pathid: 0
Refresh Epoch 5
64502, (Received from a RR-client)
  10.255.255.1 (metric 200) from 10.255.255.1 (10.255.255.1)
    Origin IGP, metric 0, localpref 100, valid, internal, best
    mpls labels in/out nlabel/10001
    rx pathid: 0, tx pathid: 0x0
ISP01-RR-R8#
```

Мы видим, что маршрут валиден, нормально резолвится через 10.255.255.1. Давайте глянем на то, что видно в FIB:

R8:

```
ISP01-RR-R8#show ip cef 172.16.255.9/32 detail
172.16.255.9/32, epoch 0, flags [rib defined all labels]
  recursive via 10.255.255.1 label 10001
    nexthop 10.2.8.2 Ethernet0/1 label 20005-(local:80007)
ISP01-RR-R8#
```

Здесь мы видим, что используется стек меток: внешняя метка - это метка next-hop'a 10.255.255.1, а внутренняя - это метка, полученная по BGP вместе с префиксом. Т.о., в LSP R8 > R16 используется пара меток на участке R8 - R1, а после выхода с R1 остается только одна метка. Проверим связность:

R8:

```
ISP01-RR-R8#ping 172.16.255.16 source Lo0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.255.16, timeout is 2 seconds:
Packet sent with a source address of 10.255.255.8
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/2 ms
ISP01-RR-R8#traceroute 172.16.255.16 source Lo0
Type escape sequence to abort.
Tracing the route to 172.16.255.16
VRF info: (vrf in name/id, vrf out name/id)
  1 10.2.8.2 [MPLS: Labels 20005/10016 Exp 0] 3 msec 2 msec 3 msec
  2 10.1.2.1 [MPLS: Labels 0/10016 Exp 0] 3 msec 2 msec 2 msec
  3 192.0.2.1 [MPLS: Label 90003 Exp 0] 3 msec 3 msec 3 msec
  4 172.31.109.10 [MPLS: Label 100005 Exp 0] 3 msec 2 msec 3 msec
```

```
5 172.31.116.16 2 msec * 3 msec
ISP01-RR-R8#
```

Собственно, трейс подтверждает вышесказанное. На R1 происходит своп метки 10016 на 90003 и отдается в сторону R9:

R1:

```
ISP01-ASBR-R1#show mpls forwarding-table labels 10016
Local      Outgoing  Prefix          Bytes Label  Outgoing  Next Hop
Label      Label     or Tunnel Id    Switched     interface
10016      90003     172.16.255.16/32 2380         Et0/3       192.0.2.1
ISP01-ASBR-R1#
```

Настройка - VPNv4 Unicast RR <> RR

На данный момент у нас есть связность между рефлекторами, поэтому приступаем к настройке eBGP-сессий в VPNv4 Unicast.

R7 / R8:

```
router bgp 64501
 neighbor RR_OPT peer-group
 neighbor RR_OPT remote-as 64502
 neighbor RR_OPT update-source Loopback0
 neighbor RR_OPT ebgp-multihop 255
 neighbor 172.16.255.15 peer-group RR_OPT
 neighbor 172.16.255.16 peer-group RR_OPT
 !
 address-family vpnv4 unicast
  neighbor 172.16.255.15 activate
  neighbor 172.16.255.16 activate
  neighbor RR_OPT send-community both
 !
 !
```

R15 / R16:

```
router bgp 64502
 neighbor RR_OPT peer-group
 neighbor RR_OPT remote-as 64501
 neighbor RR_OPT update-source Loopback0
 neighbor RR_OPT ebgp-multihop 255
 neighbor 10.255.255.7 peer-group RR_OPT
 neighbor 10.255.255.8 peer-group RR_OPT
 !
 address-family vpnv4 unicast
```

```

neighbor 10.255.255.7 activate
neighbor 10.255.255.8 activate
neighbor RR_OPT send-community both
!
!
```

Проверим на R8:

R8:

```

ISP01-RR-R8#show bgp vpnv4 unicast all summary | i 64502
172.16.255.15  4      64502      16      17      142      0      0 00:00:47
13
172.16.255.16  4      64502      17      19      142      0      0 00:00:35
13
ISP01-RR-R8#show bgp vpnv4 unicast all neighbors 172.16.255.15 routes | b Netw
      Network      Next Hop      Metric LocPrf Weight Path
Route Distinguisher: 172.16.255.9:11
*>  10.0.0.9/32      172.16.255.15      0 64502 ?
Route Distinguisher: 172.16.255.9:10001
*>  10.9.22.0/24      172.16.255.15      0 64502 ?
*>  192.168.255.22/32
      172.16.255.15      0 64502 65001 i
Route Distinguisher: 172.16.255.10:11
*>  10.0.0.10/32      172.16.255.15      0 64502 ?
Route Distinguisher: 172.16.255.11:11
*>  10.0.0.11/32      172.16.255.15      0 64502 ?
Route Distinguisher: 172.16.255.12:11
*>  10.0.0.12/32      172.16.255.15      0 64502 ?
Route Distinguisher: 172.16.255.12:10002
*>  10.112.120.0/24    172.16.255.15      0 64502 ?
*>  172.16.0.20/32     172.16.255.15      0 64502 ?
Route Distinguisher: 172.16.255.13:11
*>  10.0.0.13/32      172.16.255.15      0 64502 ?
Route Distinguisher: 172.16.255.14:11
*>  10.0.0.14/32      172.16.255.15      0 64502 ?
Route Distinguisher: 172.16.255.14:10001
*>  10.14.18.0/24      172.16.255.15      0 64502 ?
*>  192.168.255.18/32
      172.16.255.15      0 64502 65001 i
Route Distinguisher: 172.16.255.15:11
*>  10.0.0.15/32      172.16.255.15      0      0 64502 ?

Total number of prefixes 13
ISP01-RR-R8#show bgp vpnv4 unicast all 192.168.255.18/32
BGP routing table entry for 172.16.255.14:10001:192.168.255.18/32, version 140
Paths: (3 available, best #3, no table)
  Advertised to update-groups:
    6          7          8
  Refresh Epoch 2
  64502 65001
    172.16.255.16 (metric 200) (via default) from 172.16.255.16 (172.16.255.16)
```

```
Origin IGP, localpref 100, valid, external
Extended Community: RT:64502:10001
mpls labels in/out 80030/160001
rx pathid: 0, tx pathid: 0
Refresh Epoch 1
64502 65001
172.16.255.16 (metric 200) (via default) from 10.255.255.7 (10.255.255.7)
Origin IGP, metric 0, localpref 100, valid, internal
Extended Community: RT:64502:10001
mpls labels in/out 80030/160001
rx pathid: 0, tx pathid: 0
Refresh Epoch 1
64502 65001
172.16.255.15 (metric 200) (via default) from 172.16.255.15 (172.16.255.15)
Origin IGP, localpref 100, valid, external, best
Extended Community: RT:64502:10001
mpls labels in/out 80030/150019
rx pathid: 0, tx pathid: 0x0
ISP01-RR-R8#
```

Control-plane у нас сошелся, проверим теперь работу data plane. Для этого сначала для клиентских vrf добавим соответствующие RT из AS64502.

R4:

```
vrf definition CUST2
!
address-family ipv4
route-target export 64502:10002
route-target import 64502:10002
exit-address-family
!
```

R6:

```
vrf definition CUST1
!
address-family ipv4
route-target export 64502:10001
route-target import 64502:10001
exit-address-family
!
```

R8:

```
vrf definition CUST1
!
```

```

address-family ipv4
  route-target export 64502:10001
  route-target import 64502:10001
exit-address-family
!
```

Для начала посмотрим что у нас есть на PE на примере vrf CUST2, на R4:

R4:

```

ISP01-R4#show ip route vrf CUST2 | b Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.104.119.0/24 is directly connected, Ethernet1/3
L       10.104.119.104/32 is directly connected, Ethernet1/3
B       10.112.120.0/24 [200/0] via 172.16.255.16, 00:14:55
      172.16.0.0/32 is subnetted, 2 subnets
O       172.16.0.19 [110/11] via 10.104.119.19, 03:52:04, Ethernet1/3
B       172.16.0.20 [200/0] via 172.16.255.16, 00:14:55
ISP01-R4#
ISP01-R4#show bgp vpnv4 unicast vrf CUST2 | b Network
      Network          Next Hop          Metric LocPrf Weight Path
Route Distinguisher: 10.255.255.4:1002 (default for vrf CUST2)
*>  10.104.119.0/24  0.0.0.0          0          32768 ?
*>i  10.112.120.0/24  172.16.255.16     0    100      0 64502 ?
*>  172.16.0.19/32   10.104.119.19     11          32768 ?
*>i  172.16.0.20/32   172.16.255.16     0    100      0 64502 ?
ISP01-R4#
ISP01-R4#show bgp vpnv4 unicast vrf CUST2 172.16.0.20/32
BGP routing table entry for 10.255.255.4:1002:172.16.0.20/32, version 105
Paths: (1 available, best #1, table CUST2)
Flag: 0x100
Not advertised to any peer
Refresh Epoch 3
64502, imported path from 172.16.255.12:10002:172.16.0.20/32 (global)
172.16.255.16 (metric 200) (via default) from 10.255.255.7 (10.255.255.7)
Origin incomplete, metric 0, localpref 100, valid, internal, best
Extended Community: RT:64502:10002 OSPF DOMAIN ID:0x0005:0x000027120200
  OSPF RT:0.0.0.0:2:0 OSPF ROUTER ID:10.112.120.12:0
  mpls labels in/out nlabel/160011
  rx pathid: 0, tx pathid: 0x0
ISP01-R4#
```

Здесь мы видим, что маршруты были получены от наших RR, next-hop'ом считается 172.16.255.16, VPN-метка 160011.

Давайте попробуем проверить связность между CPE02-R19 и CPE02-R20:

R19:

```

CPE02-R19#ping 172.16.0.20 source Lo0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.0.20, timeout is 2 seconds:
Packet sent with a source address of 172.16.0.19
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/4/5 ms
CPE02-R19#traceroute 172.16.0.20 source Lo0
Type escape sequence to abort.
Tracing the route to 172.16.0.20
VRF info: (vrf in name/id, vrf out name/id)
 1 10.104.119.104 1 msec 0 msec 1 msec
 2 10.2.4.2 [MPLS: Labels 20005/10016/160011 Exp 0] 5 msec 4 msec 4 msec
 3 10.1.2.1 [MPLS: Labels 0/10016/160011 Exp 0] 3 msec 3 msec 4 msec
 4 192.0.2.1 [MPLS: Labels 90003/160011 Exp 0] 4 msec 4 msec 4 msec
 5 172.31.109.10 [MPLS: Labels 100005/160011 Exp 0] 4 msec 3 msec 3 msec
 6 172.31.116.16 [MPLS: Labels 0/160011 Exp 0] 4 msec 4 msec 3 msec
 7 10.112.120.12 [MPLS: Labels 0/120001 Exp 0] 3 msec 2 msec 3 msec
 8 10.112.120.20 3 msec * 4 msec

```

Связность есть. Давайте рассмотрим трейс поподробней:

№	Запись	Комментарий
1	10.104.119.104	Пакет от CPE02-R19 до ISP01-R4
2	10.2.4.2 [MPLS: Labels 20005/10016/160011 Exp 0]	Пакет от ISP01-R4 до ISP01-R2: - PUSH на R4 метку 160011 – VPN-метка префикса 172.16.0.20/32 - PUSH на R4 метку 10016 – метка префикса 172.16.255.16/32 - PUSH на R4 метку 20005 – метка префикса 10.255.255.1
3	10.1.2.1 [MPLS: Labels 0/10016/160011 Exp 0]	Пакет от ISP01-R2 до ISP01-ASBR-R1: - SWAP на R2 метки 20005 на 0 (exp-null)
4	192.0.2.1 [MPLS: Labels 90003/160011 Exp 0]	Пакет от ISP01-ASBR-R1 до ISP02-ASBR-R9: - POP на R1 метки 0 - SWAP метки 10016 на 90003
5	172.31.109.10 [MPLS: Labels 100005/160011 Exp 0]	Пакет от ISP02-ASBR-R9 до ISP02-R10: - SWAP на R9 метки 90003 на 100005
6	172.31.116.16 [MPLS: Labels 0/160011 Exp 0]	Пакет от ISP02-R10 до ISP02-RR-R16: - SWAP на R10 метки 100005 на 0 (exp-null)
7	10.112.120.12 [MPLS: Labels 0/120001 Exp 0]	Пакет от ISP02-RR-R16 до ISP02-R12: - POP на R16 метки 0 - SWAP на R16 метки 160011 на 120001 - PUSH на R16 метки 0 (exp-null)

№	Запись	Комментарий
8	10.112.120.20	Пакет от ISP02-R12 до CPE02-R20: - POP на R12 метки 0 - POP на R12 метки 120001

Собственно, из трейса мы видим, что трафик до R20 проходит через ISP02-RR-R16. Визуально, конечно, это выглядит субоптимальным путем, однако, вполне возможно, что PE R12 действительно доступен через RR. На всякий случай сделаем трейс от R4 до R12:

R4:

```
ISP01-R4#traceroute 172.16.255.12 source Loopback0
Type escape sequence to abort.
Tracing the route to 172.16.255.12
VRF info: (vrf in name/id, vrf out name/id)
 1 10.2.4.2 [MPLS: Labels 20005/10012 Exp 0] 2 msec 2 msec 1 msec
 2 10.1.2.1 [MPLS: Labels 0/10012 Exp 0] 2 msec 2 msec 2 msec
 3 192.0.2.1 [MPLS: Label 90007 Exp 0] 2 msec 2 msec 2 msec
 4 172.31.109.10 [MPLS: Label 100009 Exp 0] 2 msec 2 msec 2 msec
 5 172.31.112.12 3 msec * 3 msec
ISP01-R4#
```

По трейсу видно, что до ISP02-R12 трафик проходит напрямую от R10, не проходя через ISP02-RR-R16.

Давайте разбираться, почему так получается. Сначала посмотрим на R4:

R4:

```
ISP01-R4#show bgp vpnv4 unicast vrf CUST2 172.16.0.20/32
BGP routing table entry for 10.255.255.4:1002:172.16.0.20/32, version 105
Paths: (1 available, best #1, table CUST2)
Flag: 0x100
Not advertised to any peer
Refresh Epoch 3
64502, imported path from 172.16.255.12:10002:172.16.0.20/32 (global)
172.16.255.16 (metric 200) (via default) from 10.255.255.7 (10.255.255.7)
Origin incomplete, metric 0, localpref 100, valid, internal, best
Extended Community: RT:64502:10002 OSPF DOMAIN ID:0x0005:0x000027120200
OSPF RT:0.0.0.0:2:0 OSPF ROUTER ID:10.112.120.12:0
mpls labels in/out nolabel/160011
rx pathid: 0, tx pathid: 0x0
ISP01-R4#
```

Здесь мы видим, что маршрут мы получили от нашего RR 10.255.255.7, в качестве next-hop стоит 172.16.255.16. Почему? А потому что у нас созданы eBGP VPNv4 сессии между RR. Т.к. это eBGP, то по умолчанию в качестве next-hop проставляется адрес соседа, от которого эти маршруты получены (в нашем случае это 172.16.255.15 и 172.16.255.16).

В общем-то, на данный момент все работает, но, все же, лучше привести к нормальному виду. Для это нам необходимо на RR включить команду **next-hop-unchanged** в сторону RR из соседней AS (ну и не забыть переанонсировать маршруты).

R7 / R8:

```
router bgp 64501
!
address-family vpnv4
 neighbor RR_OPT next-hop-unchanged
exit-address-family
!
!
clear bgp vpnv4 unicast * out
```

R15 / R16:

```
router bgp 64501
!
address-family vpnv4
 neighbor RR_OPT next-hop-unchanged
exit-address-family
!
!
clear bgp vpnv4 unicast * out
```

После того, как мы выполнили эти настройки, проверим, что сейчас видно на R4:

R4:

```
ISP01-R4#show bgp vpnv4 unicast vrf CUST2 | b Network
      Network      Next Hop      Metric LocPrf Weight Path
Route Distinguisher: 10.255.255.4:1002 (default for vrf CUST2)
*>  10.104.119.0/24  0.0.0.0              0          32768 ?
*>i  10.112.120.0/24  172.16.255.12         0        100      0 64502 ?
*>  172.16.0.19/32   10.104.119.19        11          32768 ?
*>i  172.16.0.20/32   172.16.255.12         0        100      0 64502 ?
ISP01-R4#
```

Сейчас у нас в качестве next-hop'ов стоит R12, чего мы и добивались. Проверим прохождение трафика еще и трейсом с CPE02-R19:

R19:

```
CPE02-R19#traceroute 172.16.0.20 source Lo0
Type escape sequence to abort.
```

```
Tracing the route to 172.16.0.20
VRF info: (vrf in name/id, vrf out name/id)
  1 10.104.119.104 2 msec 35 msec 1 msec
  2 10.2.4.2 [MPLS: Labels 20005/10012/120001 Exp 0] 8 msec 17 msec 18 msec
  3 10.1.2.1 [MPLS: Labels 0/10012/120001 Exp 0] 19 msec 8 msec 8 msec
  4 192.0.2.1 [MPLS: Labels 90007/120001 Exp 0] 8 msec 7 msec 5 msec
  5 172.31.109.10 [MPLS: Labels 100009/120001 Exp 0] 4 msec 3 msec 2 msec
  6 10.112.120.12 [MPLS: Labels 0/120001 Exp 0] 3 msec 2 msec 2 msec
  7 10.112.120.20 3 msec * 9 msec
CPE02-R19#
```

На данный момент трафик на R12 идет напрямую с R10, т.о. мы получили оптимальный путь прохождения трафика.

Примененная конфигурация

AS64501:

#ISP01-ASBR-R1:

```
router bgp 64501
 neighbor 192.0.2.1 remote-as 64502
 neighbor 192.0.2.1 update-source Ethernet0/3
 !
 address-family ipv4
  network 10.255.255.1 mask 255.255.255.255
  neighbor 10.255.255.7 activate
  neighbor 10.255.255.8 activate
  neighbor 192.0.2.1 activate
  neighbor 192.0.2.1 send-community
  neighbor 192.0.2.1 send-label explicit-null
  neighbor RR next-hop-self
  neighbor RR send-community
  neighbor RR send-label explicit-null
 exit-address-family
 !
 !
```

#ISP01-R2:

```
router bgp 64501
 address-family ipv4
  network 10.255.255.2 mask 255.255.255.255
  neighbor 10.255.255.7 activate
  neighbor 10.255.255.8 activate
  neighbor RR send-community
  neighbor RR send-label explicit-null
 exit-address-family
```

```
!  
!
```

#ISP01-R3:

```
router bgp 64501  
  address-family ipv4  
    network 10.255.255.3 mask 255.255.255.255  
    neighbor 10.255.255.7 activate  
    neighbor 10.255.255.8 activate  
    neighbor RR send-community  
    neighbor RR send-label explicit-null  
  exit-address-family  
!  
!
```

#ISP01-R4:

```
vrf definition CUST2  
!  
  address-family ipv4  
    route-target export 64502:10002  
    route-target import 64502:10002  
  exit-address-family  
!  
router bgp 64501  
  address-family ipv4  
    network 10.255.255.4 mask 255.255.255.255  
    neighbor 10.255.255.7 activate  
    neighbor 10.255.255.8 activate  
    neighbor RR send-community  
    neighbor RR send-label explicit-null  
  exit-address-family  
!  
!
```

#ISP01-R5:

```
router bgp 64501  
  address-family ipv4  
    network 10.255.255.5 mask 255.255.255.255  
    neighbor 10.255.255.7 activate  
    neighbor 10.255.255.8 activate  
    neighbor RR send-community  
    neighbor RR send-label explicit-null  
  exit-address-family
```

```
!  
!
```

#ISP01-R6:

```
vrf definition CUST1  
!  
  address-family ipv4  
    route-target export 64502:10001  
    route-target import 64502:10001  
  exit-address-family  
!  
router bgp 64501  
  address-family ipv4  
    network 10.255.255.6 mask 255.255.255.255  
    neighbor 10.255.255.7 activate  
    neighbor 10.255.255.8 activate  
    neighbor RR send-community  
    neighbor RR send-label explicit-null  
  exit-address-family  
!  
!
```

#ISP-RR-R7:

```
router bgp 64501  
  neighbor RR_OPT peer-group  
  neighbor RR_OPT remote-as 64502  
  neighbor RR_OPT update-source Loopback0  
  neighbor RR_OPT ebgp-multihop 255  
  neighbor 172.16.255.15 peer-group RR_OPT  
  neighbor 172.16.255.16 peer-group RR_OPT  
  !  
  address-family ipv4  
    network 10.255.255.7 mask 255.255.255.255  
    neighbor 10.255.255.1 activate  
    neighbor 10.255.255.2 activate  
    neighbor 10.255.255.3 activate  
    neighbor 10.255.255.4 activate  
    neighbor 10.255.255.5 activate  
    neighbor 10.255.255.6 activate  
    neighbor 10.255.255.8 activate  
    neighbor OWN_RRC send-community  
    neighbor OWN_RRC send-label explicit-null  
    neighbor OWN_RRC route-reflector-client  
    neighbor RRN send-community  
    neighbor RRN send-label explicit-null  
  exit-address-family  
  !
```

```
address-family vpnv4 unicast
neighbor 172.16.255.15 activate
neighbor 172.16.255.16 activate
neighbor RR_OPT send-community both
neighbor RR_OPT next-hop-unchanged
!
!
```

#ISP-RR-R8:

```
vrf definition CUST1
!
address-family ipv4
route-target export 64502:10001
route-target import 64502:10001
exit-address-family
!
router bgp 64501
neighbor RR_OPT peer-group
neighbor RR_OPT remote-as 64502
neighbor RR_OPT update-source Loopback0
neighbor RR_OPT ebgp-multihop 255
neighbor 172.16.255.15 peer-group RR_OPT
neighbor 172.16.255.16 peer-group RR_OPT
!
address-family ipv4
network 10.255.255.8 mask 255.255.255.255
neighbor 10.255.255.1 activate
neighbor 10.255.255.2 activate
neighbor 10.255.255.3 activate
neighbor 10.255.255.4 activate
neighbor 10.255.255.5 activate
neighbor 10.255.255.6 activate
neighbor 10.255.255.7 activate
neighbor OWN_RRC send-community
neighbor OWN_RRC send-label explicit-null
neighbor OWN_RRC route-reflector-client
neighbor RRN send-community
neighbor RRN send-label explicit-null
exit-address-family
!
address-family vpnv4 unicast
neighbor 172.16.255.15 activate
neighbor 172.16.255.16 activate
neighbor RR_OPT send-community both
neighbor RR_OPT next-hop-unchanged
!
!
```

AS64502:

#ISP02-ASBR-R09:

```
router bgp 64502
 neighbor 192.0.2.0 remote-as 64501
 neighbor 192.0.2.0 update-source Ethernet0/3
 !
 address-family ipv4
  network 172.16.255.9 mask 255.255.255.255
  neighbor 172.16.255.15 activate
  neighbor 172.16.255.16 activate
  neighbor 192.0.2.0 activate
  neighbor 192.0.2.0 send-community
  neighbor 192.0.2.0 send-label explicit-null
  neighbor RR send-community
  neighbor RR send-label explicit-null
  neighbor RR next-hop-self
 exit-address-family
 !
 !
```

#ISP02-R10:

```
router bgp 64502
 address-family ipv4
  network 172.16.255.10 mask 255.255.255.255
  neighbor 172.16.255.15 activate
  neighbor 172.16.255.16 activate
  neighbor RR send-community
  neighbor RR send-label explicit-null
 exit-address-family
 !
 !
```

#ISP02-R11:

```
router bgp 64502
 address-family ipv4
  network 172.16.255.11 mask 255.255.255.255
  neighbor 172.16.255.15 activate
  neighbor 172.16.255.16 activate
  neighbor RR send-community
  neighbor RR send-label explicit-null
 exit-address-family
 !
 !
```

#ISP02-R12:

```
router bgp 64502
 address-family ipv4
  network 172.16.255.12 mask 255.255.255.255
  neighbor 172.16.255.15 activate
  neighbor 172.16.255.16 activate
  neighbor RR send-community
  neighbor RR send-label explicit-null
 exit-address-family
!
```

#ISP02-R13:

```
router bgp 64502
 address-family ipv4
  network 172.16.255.13 mask 255.255.255.255
  neighbor 172.16.255.15 activate
  neighbor 172.16.255.16 activate
  neighbor RR send-community
  neighbor RR send-label explicit-null
 exit-address-family
!
```

#ISP02-R14:

```
router bgp 64502
 address-family ipv4
  network 172.16.255.14 mask 255.255.255.255
  neighbor 172.16.255.15 activate
  neighbor 172.16.255.16 activate
  neighbor RR send-community
  neighbor RR send-label explicit-null
 exit-address-family
!
```

#ISP02-RR-R15:

```
router bgp 64502
 neighbor RR_OPT peer-group
 neighbor RR_OPT remote-as 64501
 neighbor RR_OPT update-source Loopback0
 neighbor RR_OPT ebgp-multihop 255
 neighbor 10.255.255.7 peer-group RR_OPT
 neighbor 10.255.255.8 peer-group RR_OPT
```

```
!  
address-family ipv4  
  network 172.16.255.15 mask 255.255.255.255  
  neighbor 172.16.255.9 activate  
  neighbor 172.16.255.10 activate  
  neighbor 172.16.255.11 activate  
  neighbor 172.16.255.12 activate  
  neighbor 172.16.255.13 activate  
  neighbor 172.16.255.14 activate  
  neighbor OWN_RRC send-community  
  neighbor OWN_RRC send-label explicit-null  
  neighbor OWN_RRC route-reflector-client  
exit-address-family  
!  
address-family vpnv4 unicast  
  neighbor 10.255.255.7 activate  
  neighbor 10.255.255.8 activate  
  neighbor RR_OPT send-community both  
  neighbor RR_OPT next-hop-unchanged  
!  
!
```

#ISP02-RR-R16:

```
router bgp 64502  
  neighbor RR_OPT peer-group  
  neighbor RR_OPT remote-as 64501  
  neighbor RR_OPT update-source Loopback0  
  neighbor RR_OPT ebgp-multihop 255  
  neighbor 10.255.255.7 peer-group RR_OPT  
  neighbor 10.255.255.8 peer-group RR_OPT  
  !  
  address-family ipv4  
    network 172.16.255.16 mask 255.255.255.255  
    neighbor 172.16.255.9 activate  
    neighbor 172.16.255.10 activate  
    neighbor 172.16.255.11 activate  
    neighbor 172.16.255.12 activate  
    neighbor 172.16.255.13 activate  
    neighbor 172.16.255.14 activate  
    neighbor OWN_RRC send-community  
    neighbor OWN_RRC send-label explicit-null  
    neighbor OWN_RRC route-reflector-client  
  exit-address-family  
  !  
  address-family vpnv4 unicast  
    neighbor 10.255.255.7 activate  
    neighbor 10.255.255.8 activate  
    neighbor RR_OPT send-community both  
    neighbor RR_OPT next-hop-unchanged
```

```
!  
!
```